

## The formaldehyde cliff lands; the silicate opening moves to inorganic ground

*With the EU formaldehyde limit halving on 6 August 2026, the shift beyond petrochemical binders is real but selective: wood-panel makers are choosing fossil MDI and lignin over silicate, while sodium-silicate chemistry gains its clearest ground in low-carbon construction, fire protection and foundry binders.*

Edition 2 · previous cycle: 'Formaldehyde cliff hardens; wood-panel substitution chooses pMDI and lignin, not silicate' (30 May 2026)

The consensus read on sustainable adhesives is that tightening chemistry rules open a wide substitution window, with silicate binders among the chemistries expected to gain. This cycle's evidence is sharper. The EU formaldehyde limit halves to 0.062 mg/m<sup>3</sup> on 6 August 2026, yet the wood-panel industry is visibly choosing fossil no-added-formaldehyde MDI and lignin, not silicate. Since the 30 May cycle the broad REACH revision has been shelved for targeted rules, softening the wider tailwind, while a soda-ash and energy squeeze has landed on European inorganic producers. The question for Bollerup Jensen: press silicate where it wins on inorganic performance, or contest a panel market already closing?

### Trend Radar

The momentum register mapped onto the horizon: ring = when a trend bites, sector = the cycle's pattern cluster, blip size = heat, arrows = momentum, dashed ghosts and tails = drift since the 30 May register. Hover any blip for its full entry; expand for the reading view.

MOMENTUM THIS CYCLE

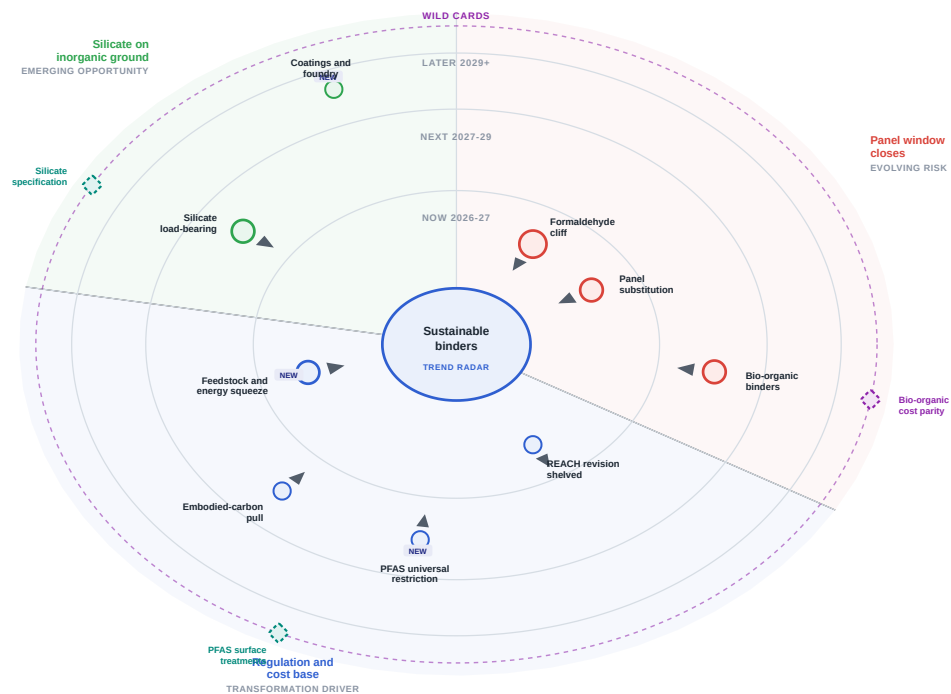
**7 Accelerating**

**1 Stable**

**1 Fading**

· 3 new to the register

● evolving risk    ● transformation driver    ● emerging opportunity    ◆ wild card    heat: ●●● cool → very hot  
 ▲ toward centre = accelerating · ▽ outward = fading    ○ drift tail = last cycle's state    ○ × left the register



## How to read this radar

- **Rings are time.** Now 2026-27 · Next 2027-29 · Later 2029+. The closer to the centre, the sooner it bites.
- ● ● **Sectors are the cycle's patterns.** Red = evolving risk, blue = transformation driver, green = emerging opportunity, the same palette as our Futures Wheel.
- **Size is heat.** Bigger blips carry higher impact × likelihood in the momentum table.
- ▲ **Arrows are momentum.** Pointing toward the centre = accelerating; outward = fading. No arrow = stable.
- ... **Dashed ghosts are last cycle.** The dotted tail shows the drift since the 30 May register.
- ◆ **Diamonds on the outer rim are wild cards,** low-probability, high-impact discontinuities, each with its tripwires.

**Hover any blip for its full register entry, and click to pin it. This guide returns when you move away, or press ? any time.**

## Momentum across the topic

| Trend                                             | Cluster               | Horizon   | Heat     | Momentum     | Evidence                                                                                                                       |
|---------------------------------------------------|-----------------------|-----------|----------|--------------|--------------------------------------------------------------------------------------------------------------------------------|
| EU formaldehyde cliff now imminent                | EVOLVING RISK         | 2026      | VERY HOT | ACCELERATING | Named 6 August 2026 application date, weeks from the cycle; 3 sources within 120 days ( <a href="#">Notifix</a> , April 2026). |
| Panel substitution locks to fossil MDI and lignin | EVOLVING RISK         | 2026-2027 | HOT      | ACCELERATING | MEDITE, Egger and Koskisen moves; 3 sources April to June 2026 ( <a href="#">Housebuilder &amp; Developer</a> , April 2026).   |
| Bio-organic binders take the near-term path       | EVOLVING RISK         | 2026-2029 | HOT      | ACCELERATING | Peer review plus lignin product expansion; sources May to July 2026 ( <a href="#">Polymers</a> , July 2026).                   |
| REACH broad revision shelved for targeted rules   | TRANSFORMATION DRIVER | 2026-2027 | MODERATE | FADING       | Commission decision reported May; distribution even across the window ( <a href="#">SOCI</a> , May 2026).                      |

| Trend                                                          | Cluster               | Horizon   | Heat     | Momentum     | Evidence                                                                                                                        |
|----------------------------------------------------------------|-----------------------|-----------|----------|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>PFAS universal restriction advances</b><br>NEW              | TRANSFORMATION DRIVER | 2026-2029 | MODERATE | ACCELERATING | ECHA SEAC draft opinion and consultation; source March 2026 ( <a href="#">Covington &amp; Burling</a> , March 2026).            |
| <b>Construction embodied-carbon pull on binders</b>            | TRANSFORMATION DRIVER | 2026-2030 | MODERATE | ACCELERATING | New EU life-cycle carbon method published; source May 2026 ( <a href="#">European Commission</a> , May 2026).                   |
| <b>Feedstock and energy squeeze inorganic producers</b><br>NEW | TRANSFORMATION DRIVER | 2026-2028 | HOT      | ACCELERATING | Solvay capacity cut plus Cefic power demand; 2 sources February 2026 ( <a href="#">Solvay</a> , February 2026).                 |
| <b>Silicate geopolymers cross into load-bearing use</b>        | EMERGING OPPORTUNITY  | 2026-2030 | HOT      | ACCELERATING | Hoffmann Green ATec plus Ecocem full-scale build; sources January and July 2026 ( <a href="#">World Cement</a> , January 2026). |

| Trend                                                                 | Cluster                | Horizon       | Heat      | Momentum | Evidence                                                                                                                               |
|-----------------------------------------------------------------------|------------------------|---------------|-----------|----------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inorganic chemistry advances in coatings and foundry</b><br>NEW    | EMERGING OPPORTUNITY   | 2026-2032     | MODERATE  | STABLE   | Coatings reformulation and a foundry-binder move; sources February and June 2026 ( <a href="#">European Coatings</a> , June 2026).     |
| <b>Silicate-activated chemistry lands a flagship EU specification</b> | DISRUPTIVE OPPORTUNITY | Wild-card rim | HIGH      | —        | A procurement or fire-safety standard writes silicate-activated binder or coating into spec language (Change Tracker, 22 July 2026).   |
| <b>Bio-organic binders reach cost parity and take the panel share</b> | DISRUPTIVE RISK        | Wild-card rim | VERY HIGH | —        | A lignin or soy binder reaches fossil-resin cost parity and captures the formaldehyde-free panel share (Change Tracker, 22 July 2026). |

| Trend                                                                    | Cluster                | Horizon       | Heat | Momentum | Evidence                                                                                                                |
|--------------------------------------------------------------------------|------------------------|---------------|------|----------|-------------------------------------------------------------------------------------------------------------------------|
| PFAS restriction turns surface treatments into the biggest silicate pull | DISRUPTIVE OPPORTUNITY | Wild-card rim | HIGH | —        | PFAS reformulation makes water-glass surface treatments the largest new silicate demand (Change Tracker, 22 July 2026). |

Ring = when the trend bites (Now 2026-27 · Next 2027-29 · Later 2029+), sector = pattern cluster, blip size and stroke = heat (impact × likelihood), arrows = momentum direction this cycle. Wild cards sit on the dashed outer rim. **Drift tails** show movement against the 30 May register: a dashed ghost is the trend’s previous state and the dotted tail leads to where it sits now. Trends with no tail held their grades; this cycle widens the topic from silicate adhesives, so the register is largely re-drawn and prior-cycle grades are not carried at signal level.

### Scenario Matrix

Two uncertainties frame Bollerup Jensen's next moves: which chemistry captures the sustainable-binder shift, and how fast the demand pull forms. The four pairings:

SUBSTITUTION VERDICT



## The Commodity Squeeze

The worst pairing: the formaldehyde cliff is met on cost, not performance, and the sustainable shift stays diffuse. Panel and coating buyers pick the cheapest compliant route, reformulated fossil MDI or lignin; no low-carbon or fire specification pays a performance premium; and silicate stays a low-margin commodity just as the soda-ash and energy squeeze compresses margins. Bollerup Jensen defends volume on price in a shrinking, undifferentiated pool.

**Indicators:** no second silicate-binder Technical Assessment lands; PFAS derogations widen or the restriction slips; tenders specify on price; Nordic power prices climb toward continental levels.

## Right Chemistry, Wrong Clock

Silicate is the right answer, but demand forms too slowly to pay off this horizon. The inorganic case in load-bearing geopolymer, fire protection and foundry cores is proven and Bollerup Jensen holds the chemistry, yet standards translate slowly, procurement lags, and adoption stays at pilot scale. The opportunity is real but sits beyond the cycle, and the wait has to be funded.

**Indicators:** silicate-activated approvals stay isolated; embodied-carbon declarations roll out but procurement does not yet price them; foundry inorganic-binder use stays at pilot; coating reformulation is announced, not specified.



### The Fast Bypass

Demand moves fast and specification-led, but around bio-organic and reformulated fossil chemistries, not silicate. Lignin and soy binders reach cost parity, PFAS reformulation flows to organic systems that are not silicate, and the quick specification wave in panels and coatings is written for chemistries Bollerup Jensen does not supply. A real shift arrives, and silicate is not on the ticket.

**Indicators:** a third major panel line converts to lignin or soy; a lignin binder reaches cost parity; a coatings major launches a non-silicate organic line; retailers specify bio-bonded panels.

### Specified In

The best pairing: the sustainable shift concentrates in inorganic-performance markets and moves at specification speed. Silicate-activated geopolymer binders cross into load-bearing use, PFAS and fire-safety rules pull surface treatments toward mineral systems, and foundries commit to inorganic cores, fast enough to write silicate into specifications. Bollerup Jensen's Nordic, low-carbon silicate is specified in as a performance input, not priced as a commodity.

**Indicators:** a second silicate-binder Technical Assessment in another EU state; a fire-protection certificate for a silicate-bonded assembly; a European foundry commits to an inorganic-only roadmap; a coatings major adopts an inorganic silicate line.

## Pattern narrative

### The panel window closes to non-silicate routes EVOLVING RISK

Three signals point the same way. The 0.062 mg/m<sup>3</sup> formaldehyde limit lands on 6 August 2026 with no grace period, a deadline Fraunhofer IBP underlines for the largest EU market ([Fraunhofer IBP](#), May 2026). The panel industry's answer is not silicate: MEDITE's compliant range runs on no-added-formaldehyde MDI, Egger has launched a formaldehyde-free bonded board ([Wood & Panel](#), June 2026), and Koskisen and Stora Enso are widening their lignin-bonded Zero range ([Wood & Panel](#), May 2026). The panel federation puts the resin-cost step at 15% to 25%. The wood-panel opening that framed the last cycle is narrowing as fossil and bio-organic routes take the contracts.

## Regulation narrows and the cost base tightens TRANSFORMATION DRIVER

The regulatory picture has shifted since May. Brussels has shelved the broad REACH revision for targeted restrictions, pruning the wider tailwind. Two rules still bite: the advancing PFAS restriction pushes fluorinated coatings toward reformulation, and the new EU building-carbon method deepens the low-carbon binder pull. Against this, a cost squeeze has appeared: Solvay is cutting Torrelavega soda ash to 420 kilotons (from 600 kt previously) and Europe's energy-intensive producers are pressing Brussels on power prices ([Cefic](#), February 2026). Nordic electricity remains Bollerup Jensen's quiet advantage to protect.

## Silicate wins on inorganic ground EMERGING OPPORTUNITY

Where silicate competes on inorganic performance, momentum is clearer. Hoffmann Green's silicate-activated 0% clinker cement now carries a load-bearing Assessment, and Ecocem has taken a low-carbon binder to a full-scale residential build ([World Cement](#), July 2026). Surface treatments and foundry chemistry add to it: coatings are reformulating toward mineral systems, and foundry-binder suppliers are positioning inorganic silicate cores ([Cast Metals Federation](#), February 2026). These are the fronts where a Danish silicate maker sells performance, not price.

## Signals gaining momentum

1. **The formaldehyde cliff lands.** On 6 August 2026 the E05 limit applies with no grace period, forcing every panel maker's binder choice within weeks.
2. **Feedstock and energy squeeze.** Solvay's soda-ash cut and industry power-cost demands signal a tightening European cost base that rewards low-energy Nordic production.
3. **PFAS restriction advances.** The SEAC draft opinion moves the universal ban forward, pulling fluorinated coating and sealant chemistries toward inorganic swaps.
4. **Silicate crosses into load-bearing use.** Certified silicate-activated cement and full-scale low-carbon builds extend the specification template silicate systems can lean on.

## Wild Cards to Watch

### Silicate-activated chemistry lands a flagship EU specification

**DISRUPTIVE OPPORTUNITY** *Potential impact: High*

**Surprise characteristics:** A national procurement or fire-safety standard writes silicate-activated binder or coating into specification language.

**Early warning indicators:** A second silicate-binder Technical Assessment in another EU state; a notified body certifies a silicate-bonded assembly.

The Hoffmann Green load-bearing Assessment is the template a silicate adhesive or coating line would lean on; a second national approval in the next two quarters would turn adjacent proof into a specification silicate producers can sell into ([World Cement](#), January 2026).

### Bio-organic binders reach cost parity and take the panel share

**DISRUPTIVE RISK** *Potential impact: Very High*

**Surprise characteristics:** A lignin or soy binder reaches fossil-resin cost parity and captures the whole formaldehyde-free panel share, closing silicate out of wood panels.

**Early warning indicators:** A third major panel line converting to lignin or soy; a tier-1 retailer specifying bio-bonded panels.

Koskisen and Stora Enso's Zero expansion shows the bio-organic route already scaling ([Wood & Panel](#), May 2026); a cost-parity breakthrough would settle the panel question against silicate for a decade.

### PFAS restriction turns surface treatments into the biggest silicate pull

**DISRUPTIVE OPPORTUNITY** *Potential impact: High*

**Surprise characteristics:** PFAS reformulation and a coatings major's inorganic pivot make water-glass surface treatments, not construction, the largest new silicate demand.

**Early warning indicators:** A coatings major announcing an inorganic silicate line; a fire-protection certificate for a silicate-bonded coating.

Coatings are already reformulating toward mineral systems ([European Coatings](#), June 2026); if the PFAS ban lands hard, silicate surface treatments could scale faster than the construction route the last cycle emphasised.

## Implications

This cycle reads the sustainable-binder shift as real but selective, and the wood-panel adhesive window as closing to non-silicate routes. Worth monitoring: the next lignin or MDI conversion by an EU panel major, and whether the Nordic power advantage holds as the energy squeeze spreads. Warrants preparation: a Bollerup Jensen position that leads with silicate where it wins on inorganic performance, in low-carbon construction binders, fire protection and foundry cores, rather than contesting panel adhesives head-on ([Bollerup Jensen](#), evergreen). A decision point opens if a second silicate-binder Technical Assessment or a PFAS-driven coating reformulation lands next quarter. The EU is backing the move beyond petrochemicals, a shift in which the EU bioeconomy was estimated at €2.7 trillion in 2023 ([Council of the European Union](#), March 2026).

## Bibliography

Source tiers: **Tier 1**, governments, regulators and intergovernmental bodies. **Tier 2**, think-tanks, academic institutes, major consultancies and quality data providers. **Tier 3**, quality journalism and specialist trade press. **Tier 4**, vendor, company and practitioner sources, used only as directional corroboration.

- **Tier 1** EU building life-cycle GWP calculation method. [European Commission \(Energy\)](#) (04/05/2026).
- **Tier 1** Council backs the EU Bioeconomy Strategy. [Council of the European Union](#) (17/03/2026).
- **Tier 2** Formaldehyde limits 2026: what companies must do. [Fraunhofer IBP](#) (20/05/2026).
- **Tier 2** Bio-based wood adhesives review. [Polymers \(MDPI\)](#) (09/07/2026).
- **Tier 2** ECHA consultation on a universal PFAS ban. [Covington & Burling](#) (27/03/2026).
- **Tier 2** Energy-intensive industries demand affordable EU power. [Cefic](#) (24/02/2026).
- **Tier 3** EU halves the wood-panel formaldehyde limit from August 2026. [Notifix](#) (23/04/2026).
- **Tier 3** MEDITE SMARTPLY MDF range compliant via NAF-MDI. [Housebuilder & Developer](#) (10/04/2026).
- **Tier 3** Egger DHF panel with formaldehyde-free bonding. [Wood & Panel](#) (30/06/2026).
- **Tier 3** Koskisen and Stora Enso expand lignin-bonded Zero panels. [Wood & Panel](#) (21/05/2026).

- **Tier 3** Europe steps back from a broad REACH revision. [SOCl](#) (06/05/2026).
- **Tier 3** Hoffmann Green H-UKR 0% clinker cement validated for superstructures. [World Cement](#) (12/01/2026).
- **Tier 3** Ecocem ACT cement reaches first full-scale residential deployment. [World Cement](#) (17/07/2026).
- **Tier 3** Sustainable coatings: from ambition to implementation. [European Coatings](#) (15/06/2026).
- **Tier 3** Huettenes-Albertus joins Cast Metals Federation on inorganic binders. [Cast Metals Federation](#) (09/02/2026).
- **Tier 4** Solvay cuts Torrelavega soda ash capacity. [Solvay](#) (23/02/2026).
- **Tier 4** Liquid sodium, potassium and lithium silicates. [Bollerup Jensen Group](#) (Evergreen vendor page).

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